

NWIS

WELL COMPLETION REPORT

Doc ID
W001-WCR-01

Well: Alpha-01 | Well ID: W001 | Field: Field_Alpha

PRIMARY ISSUE CATEGORY: MUD LOSS

1. GENERAL WELL INFORMATION

Well Name	Alpha-01	Well ID	W001
Field	Field_Alpha	Block	Block_1
Spud Date	02-05-2024	Rig Release Date	29-05-2024
Total Depth (TD)	3050 m	Rig Name	Rig Sagar-3
Well Type	Vertical Exploratory	Trajectory	Vertical
Max Inclination	2.1 deg	Directional Interval	N/A (near-vertical)

2. FORMATION & PORE PRESSURE INFORMATION

Formation_X (unconsolidated sand-shale sequence) was encountered between 2850 m and 3100 m, with normal pressure gradient of approximately 1.05-1.10 SG equivalent mud weight. The sand streak at 2880-2950 m shows high permeability (estimated 200-500 mD from offset core data) and is historically associated with losses of circulation across Field_Alpha.

3. LITHOLOGY SUMMARY

Interval (m)	Formation	Lithology	Notes
2500-2850	Formation_W	Shale, grey, firm	Normal drilling trend
2850-2880	Formation_X (upper)	Shale, interbedded thin sand	Transition zone
2880-2950	Formation_X (sand)	Sand, friable, high permeability	Primary loss zone
2950-3020	Formation_X (lower)	Sand, friable, interbedded shale	Loss-prone; monitor returns
3020-3050	Formation_Y	Shale, grey-black, hard	Normal drilling trend

4. HISTORICAL DRILLING / OPERATIONS EVENTS

Event 1: Partial Mud Loss at approximately 2905 m

Observation: Pit volume dropped by roughly 40 bbl over 45 minutes while drilling the high-permeability sand streak of Formation_X; flow-out to flow-in ratio fell to 0.65.

Engineering Action: Drilling was stopped, pumps slowed to 40 SPM, and a 35 bbl fine-to-medium LCM pill (60 lb/bbl calcium carbonate + mica blend) was spotted across 2895-2915 m. Mud weight was reduced from 1.22 SG to 1.18 SG to lower ECD.

Outcome: Losses decreased from approximately 8 bbl/hr to negligible (<1 bbl/hr) within 90 minutes. Full returns confirmed before resuming drilling; 6.5 hours of NPT recorded.

5. CASING AND COMPLETION SUMMARY

Casing	Size (in)	Set Depth (m)	Cement Top (m)
Conductor	20	60	Surface

Surface	13-3/8	650	Surface
Intermediate	9-5/8	2450	1200
Production	7	3050	2200

6. NON-PRODUCTIVE TIME (NPT) SUMMARY

Issue	Depth (m)	NPT (hrs)	Category
Partial mud loss at 2905 m	2905	6.5	Lost Circulation

7. ROOT CAUSE ANALYSIS

Root cause was identified as an unsupported, high-permeability sand streak within Formation_X combined with an ECD that briefly exceeded the formation's fracture/loss threshold during a connection. Offset well data (W003) shows an analogous loss event at a comparable structural depth, confirming this is a field-wide characteristic of the Formation_X sand rather than a localized anomaly.

8. LESSONS LEARNED

- Formation_X sand between 2880-2950 m is confirmed loss-prone across at least two wells in Field_Alpha.
- Pre-mixing 30-40 bbl of LCM pill before entering 2870 m reduces reaction time and total NPT.
- Reducing mud weight by 0.03-0.05 SG ahead of the sand streak lowers ECD enough to avoid triggering losses in most cases.
- Flow-out monitoring at higher sampling frequency (every 5 min vs 15 min) during this interval gives earlier warning.

9. RECOMMENDATION FOR OFFSET WELLS

For any future well targeting Formation_X in Field_Alpha, pre-mix LCM pills before reaching 2870 m, trim mud weight by 0.03-0.05 SG through the 2880-2950 m sand interval, and budget 6-8 hours of contingency NPT for loss control in the AFE.